IPL EDA

What Decides an IPL Match?

1. Basic Info About This Project

- **Project name:** IPL EDA (Exploratory Data Analysis)
- **Goal:** Look at IPL data from the first season to the latest season, ask clear questions, and answer them with numbers and charts.
- **Dataset source:** https://www.kaggle.com/datasets/maratheabhishek/ipl-dataset-2008-to-2025
- **Seasons covered:** 2008 to 2026 (19 seasons). The 2020 season is labelled 2020/21 in the data, and 2026 has only 43 matches.
- **Tools used:** Python, pandas, numpy, matplotlib, seaborn, Jupyter Notebook (in VS Code)
- **Author:** Jagjeet Sonwalkar

2. Why I Did This Project

IPL is one of the most watched cricket leagues, and many people have strong opinions about it. For example, people say "winning the toss is important" or "chasing is easier." I wanted to check these opinions using real data.

3. Questions I Wanted to Answer

1. Does winning the toss help, and does it change with the venue or the bat/field decision?
2. How has scoring changed from season to season?
3. Which part of the innings (powerplay, middle, death) decides matches?
4. Which venues help batters and which help bowlers?
5. Is chasing more successful now than in the early seasons?
6. Who are the best batters and bowlers?

4. About the Data

I used 5 files.

File	Rows	Columns	What it has
ball_by_ball_data.csv	288,226	30	Every ball: batter, bowler, runs, wickets, extras
ipl_matches_data.csv	1,212	23	Every match: teams, venue, toss, winner, date
players-data-updated.csv	799	6	Player names, batting style, bowling style
team_aliases.csv	46	3	Old and new names of teams
teams_data.csv	16	2	Team id and team name

Important columns:

- **ball_by_ball:** match_id, batter, bowler, team_batting, over_number, batter_runs, extras, total_runs, is_wicket, wicket_kind, is_super_over, innings

- **matches:** match_id, season, match_date, venue, city, toss_winner, toss_decision, match_winner, win_by_runs, win_by_wickets, player_of_match

5. Data Cleaning

I made copies of the data first, so the original files stay safe. All cleaning was done on the copies.

5.1 Team names

- Some teams changed their names (for example Delhi Daredevils became Delhi Capitals).
- I joined the aliases file with the teams file and gave every team one latest name.
- I added a short name for each team (MI, CSK, RCB and so on).
- I saved this table as team_names.csv.

5.2 Missing values

Column	Why it was missing	What I did
city	Not written for some venues	Filled by venue (Dubai and Sharjah)
match_number	Playoff matches have no number	Filled in order, and made an is_playoff column
win_by_runs, win_by_wickets	Match was won the other way	Filled with 0
player_of_match	Match had no result	Filled with -1
batsman_type, bowler_type	Not available	Filled with "Unknown"
player_out, fielders_involved, wicket_kind	No wicket on that ball	Filled with "none"
bowl_style, field_pos (players)	Not available	Filled with "Unknown"

5.3 Duplicates

- I checked for full duplicate rows in all files. **0 duplicates** were found.
- I did not drop balls that only shared the same over and ball number, because wides and no-balls are real deliveries.

5.4 New columns I made

- **year / season info:** made from match_date
- **has_result:** True if the match had a winner
- **toss_win_match_win:** True if the toss winner also won the match
- **chasing_team, chase_win:** the team batting second, and whether it won
- **first_inns_runs, second_inns_runs:** total runs in each innings
- **over0:** over number from 0 to 19
- **phase:** Powerplay (overs 1-6), Middle (7-15), Death (16-20)
- **is_legal_ball, is_four, is_six, is_dot, bowler_runs, is_bowler_wicket:** ball-level helpers

5.5 Dropped columns

I dropped columns that had one value in every row or that repeated other columns:

- **balls:** wide_ball_runs, no_ball_runs, leg_bye_runs, bye_runs, penalty_runs, is_penalty
- **matches:** balls_per_over, event_name, gender, match_type, format, overs, team_type
- **players:** player_full_name

5.6 Saved files

The clean data is saved in data/cleaned_data/:

- matches_clean.csv
- ball_by_ball_clean.csv
- players_clean.csv
- team_names.csv

5.7 Size before and after cleaning

File	Shape before	Shape after	Memory before (MB)	Memory after (MB)
balls	(288226, 30)	(288226, 24)	69.52	64.60
matches	(1212, 23)	(1212, 17)	0.32	0.21
players	(799, 6)	(799, 5)	0.08	0.06

6. Rules I Followed in the Analysis

- Super overs were left out of all scoring numbers.
- Matches with no result were left out of toss and chasing numbers.
- For venues, I only used venues with at least **20 matches**.
- For players, I only used players with at least **500 balls** faced or bowled.
- I compared phases **per over**, because the phases have different lengths.
- For small samples I show a **95% margin of error**, so a lucky number does not look like a real pattern.

7. Findings

Q1. Does winning the toss help?

- Toss winners won **51.7%** of **1,212** matches (95% range: 48.8% to 54.5%).
- Choosing to bat first: **45.1%** (412 matches, 95% range 40.3% to 50.0%). Choosing to field first: **55.0%** (800 matches, 95% range 51.6% to 58.5%).
- Venue result: only **2 of 16** venues (with 20 or more matches) showed a clear toss effect: Ekana (toss winners won 69.2% of 26 matches) and Rajiv Gandhi International (39.1% of 87 matches). With 16 venues, about 1 such result is expected by luck alone.
- Season result: in every season the 95% range includes 50%, so no single season shows a clear toss effect. Seasons ranged from 43.7% (2024) to 60.8% (2025), which is normal noise for 43 to 76 matches.
- **Simple answer:** Winning the toss alone gives no clear advantage. But the choice matters: toss winners who chose to field won 55.0% of their matches, while those who chose to bat won only 45.1%. This fits the chasing result in Q5.

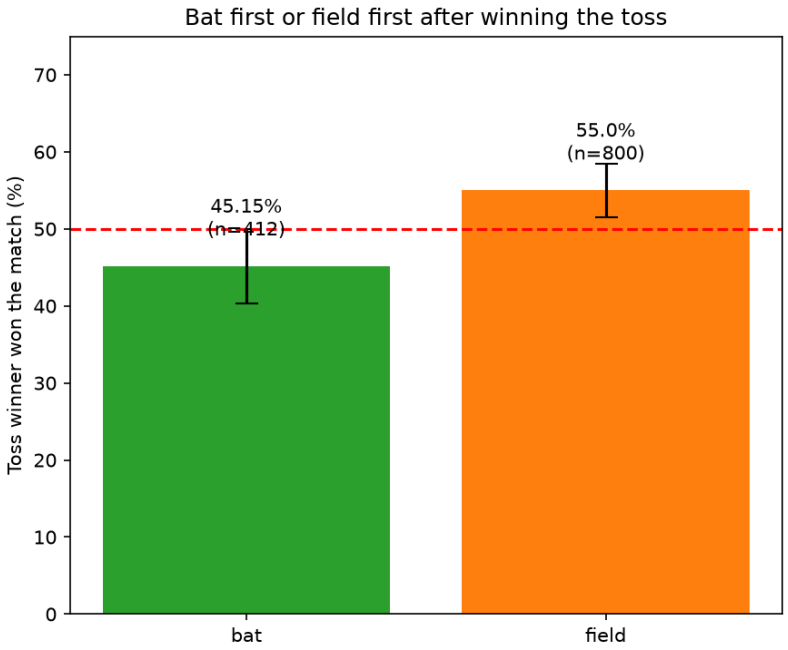


Figure 1. Toss winner win rate when choosing to bat or field (95% margin of error). The dashed line is 50%.

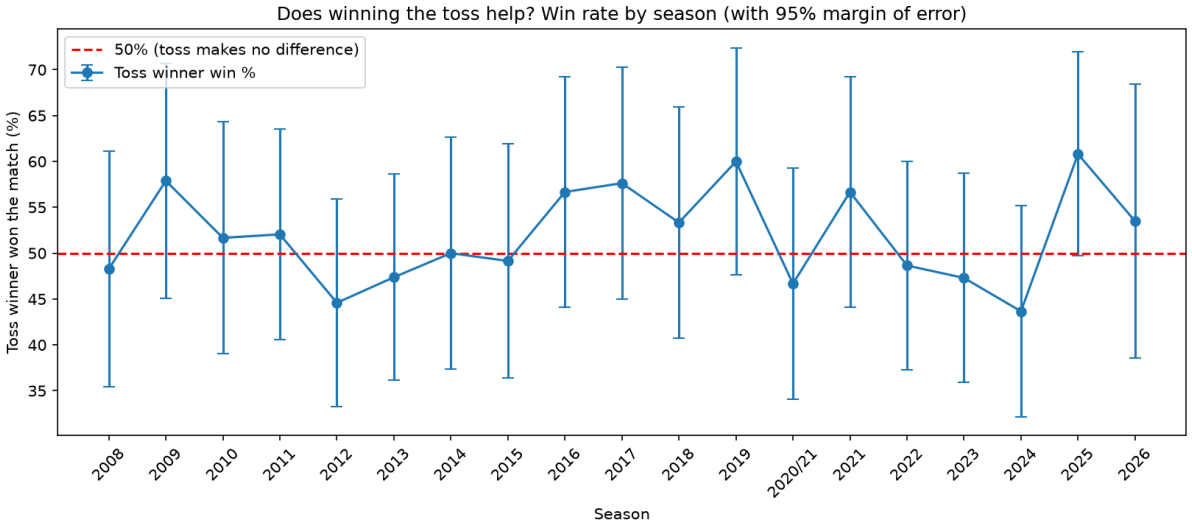


Figure 2. Toss winner win rate by season (95% margin of error). Every season range includes 50%.

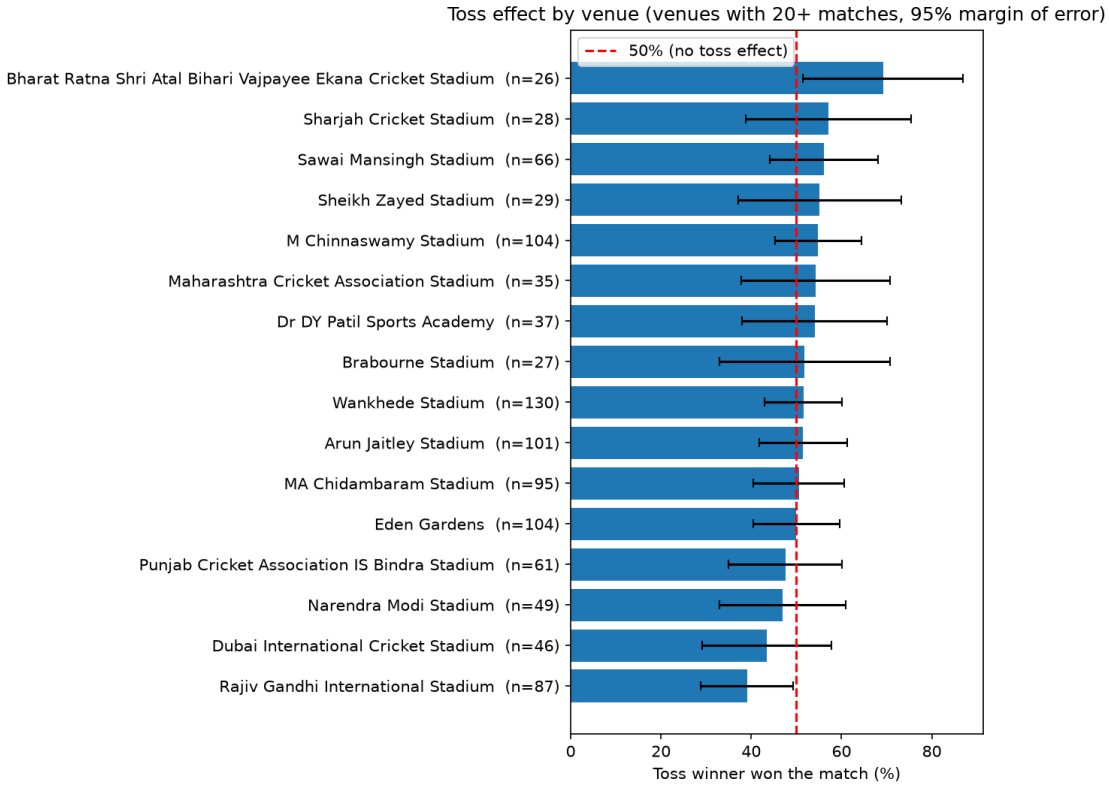


Figure 3. Toss winner win rate by venue, for venues with 20 or more matches.

Q2. How has scoring changed?

- Average first-innings score went from **161.0** in 2008 to **188.6** in 2026 (highest: 189.6 in 2024, lowest: 150.3 in 2009).
- Run rate went from **8.3** to **9.9** runs per over.
- Sixes per match went from **10.7** to **18.4**, an increase of about 72%.
- From 2008 to 2022 the average first-innings score stayed between 150 and 173. It then jumped to 182.7 in 2023 and has been about 189 since 2024.
- **Simple answer:** Scoring has gone up a lot, and most of the rise came in the last four seasons. The jump starts in 2023, the year the Impact Player rule began, but this data cannot prove that the rule caused it.

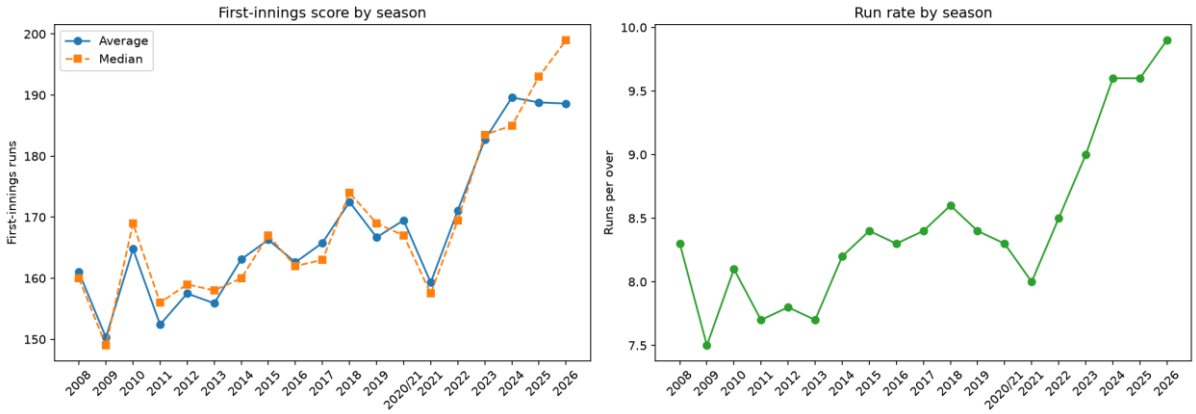


Figure 4. First-innings score (average and median) and run rate by season.

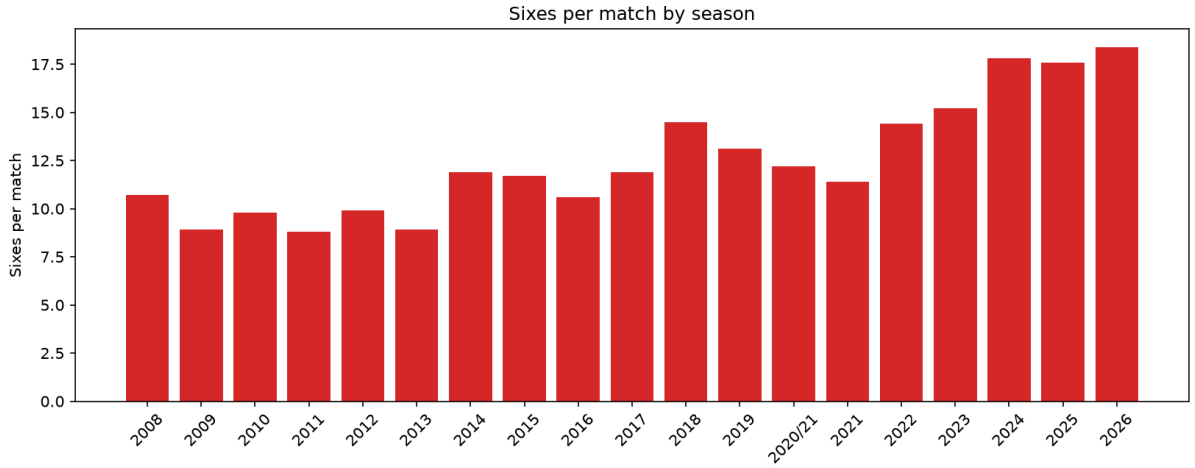


Figure 5. Sixes per match by season.

Q3. Which phase decides matches?

- Runs per over: Powerplay **8.03**, Middle **7.92**, Death **10.04**.
- Wickets per over: Powerplay **0.24**, Middle **0.26**, Death **0.51**.
- In the first innings, the phase with the biggest gap between winners and losers is **Death** (gap of **1.80** runs per over: winners 11.21, losers 9.42). Middle overs gap: 1.26. Powerplay gap: 1.03.
- Link with winning (correlation): Middle 0.32, Death 0.30, Powerplay 0.24.
- **Simple answer:** The death overs separate winners from losers the most, and they score about 2 runs per over more than the other phases while losing twice as many wickets. But the middle overs matter almost as much, so matches are won in all three phases, not only at the end.

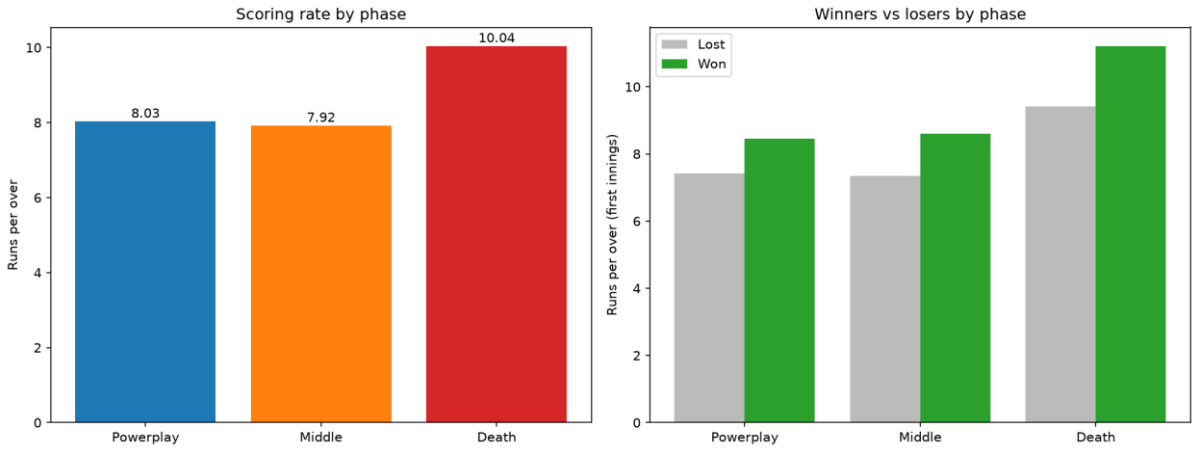


Figure 6. Scoring rate by phase (left) and first-innings winners vs losers by phase (right).

Q4. Which venues help batters or bowlers?

- Most batter-friendly venue: **Brabourne Stadium** (+0.63 runs per over above the season average, but only 25 matches). With a bigger sample: **M Chinnaswamy Stadium** (+0.36, 92 matches).
- Most bowler-friendly venue: **Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cricket Stadium** (0.86 runs per over below the season average, 25 matches). With a bigger sample: **MA Chidambaram Stadium** (0.34 below, 93 matches).
- Only **5 of 16** venues were clearly different from average: Brabourne, M Chinnaswamy, Punjab Cricket Association IS Bindra (+0.29), MA Chidambaram and Ekana.

- **Simple answer:** Most venues play close to the season average. Chinnaswamy and Mohali help batters, and Chennai and Lucknow help bowlers. Brabourne and Ekana have only 25 matches each, so treat them with care.

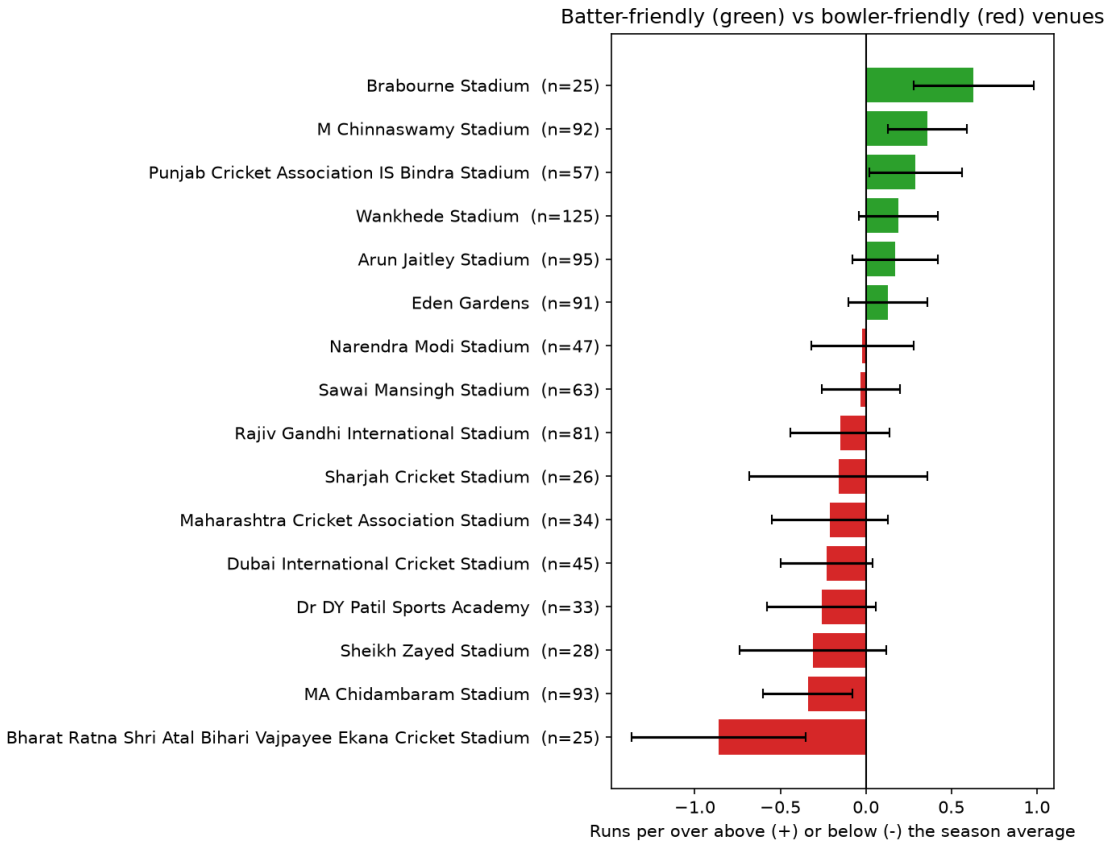


Figure 7. Runs per over above (green) or below (red) the season average, by venue. Lines show the 95% margin of error.

Q5. Is chasing more successful now?

- Overall, chasing teams won **54.7%** of matches (660 of 1,206; 95% range 51.9% to 57.5%).
- Early seasons (2008 to 2016): **54.6%**. Recent seasons (2017 to 2026): **54.8%**.
- Trend: **+0.07** percentage points per season, which is almost flat.
- Single seasons ranged from 43.1% (2015) to 68.3% (2016), which is normal noise for 60 to 75 matches.
- Teams now choose to field much more often. Of all toss winners, 800 of 1,212 (66%) chose to field. From the chart, this was roughly 35% to 55% of toss winners in 2008 to 2010, and roughly 70% to 85% since 2021.
- **Simple answer:** Chasing has been a little better than batting first (about 55%) in every part of the IPL, and it is not getting more successful over time, even though scores have risen.

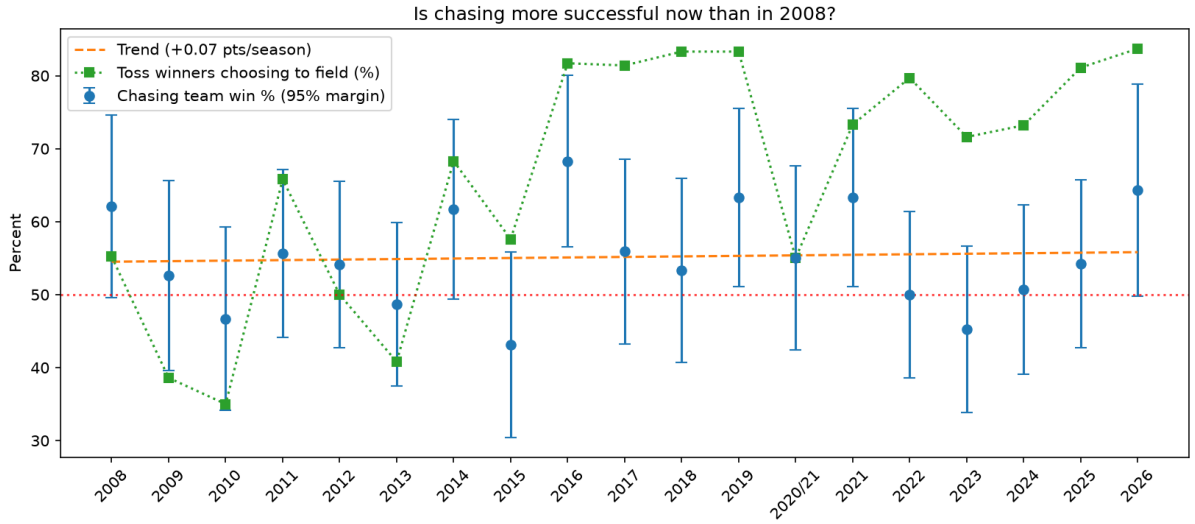


Figure 8. Chasing team win rate by season (95% margin of error), with the straight-line trend and the share of toss winners choosing to field.

Q6. Best players

- Most runs: **V Kohli**, 9,040 runs (strike rate 134.0, average 40.0). Next: RG Sharma 7,183 and S Dhawan 6,769.
- Best strike rate among the top 10 run scorers: **AB de Villiers**, 151.7. Best average among them: **KL Rahul**, 46.7.
- Most wickets: **YS Chahal**, 228 wickets (economy 8.03, average 23.1). Next: B Kumar 215 and SP Narine 199.
- Best economy among the top 10 wicket takers: **SP Narine**, 6.79. Best average and best strike rate among them: **SL Malinga**, 19.8 runs and 16.6 balls per wicket.
- Best death-over batter and bowler: **AB de Villiers** with 222.9 Best strike rate and **SP Narine** with 7.29. Best economy.

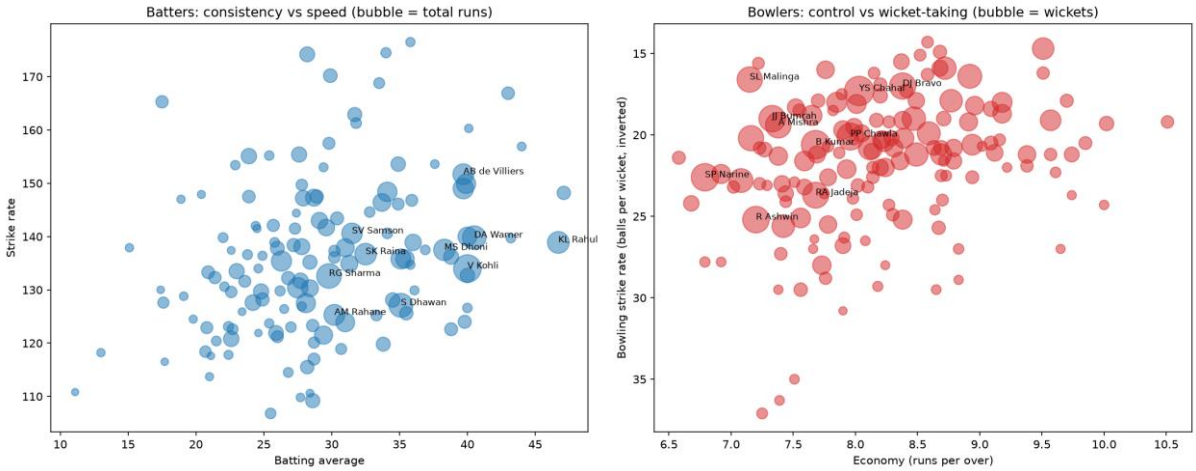


Figure 9. Batters (left, bubble = total runs) and bowlers (right, bubble = wickets). Only the top players are named.

8. Main Conclusion

The biggest change in the IPL is how much scoring has gone up: the average first-innings score rose from about 161 in 2008 to about 189 now, and sixes per match went from 10.7 to 18.4, mostly since 2023. Winning the toss alone does not give a clear edge (51.7%), but fielding first has worked better than batting first: toss winners who chose to field won 55.0% of matches, and chasing teams win

about 55% overall in every part of the IPL. Within an innings, the death overs separate winners from losers the most, although the middle overs matter almost as much. Venue effects are small, with only 5 of 16 venues clearly different from the season average.

9. Things That Surprised Me

First, winning the toss did not help by itself, but the decision did: toss winners who chose to bat won only 45.1% of their matches. Second, chasing success has stayed flat at about 55% even though scores have gone up so much and toss winners now choose to field far more often (roughly 70% to 85% of matches since 2021, against 35% to 55% in 2008 to 2010). Third, Ekana looks batter-friendly by raw run rate (8.70), but after comparing with each season it is the most bowler-friendly venue, because it was used mostly in recent high-scoring seasons.

10. Limits of This Analysis

- Each season has only about 60 to 75 matches, so single-season numbers can be noisy.
- Venue results also include pitch, weather, dew and home-team effects.
- Player numbers cover all seasons together, but scoring has changed over time.
- The results show patterns, not proof of cause.
- Rain-affected matches are included, and a few of them favour the chasing team.

11. What I Can Do Next

- Build a model to predict the match winner.
- Check the effect of the Impact Player rule from 2023.
- Compare home and away performance for each team.

12. Project Folder

```
IPL - EDA/
  data/
    (5 raw csv files)
    cleaned_data/
  images/
  outputs/
        IPL_EDA_Report
  src/
        cleaning_the_data.ipynb
        ipl_story.ipynb
```

13. How to Run This Project

1. Download the dataset from the Kaggle link and put the 5 csv files in the data folder.
2. Install the libraries: pip install pandas numpy matplotlib seaborn jupyter
3. Open the notebook in the notebooks folder.
4. Run all cells from top to bottom.